

Junhuai Xu

Ph.D. Candidate, Experimental Nuclear and Particle Physics
Ph.D. Advisor: Zhigang Xiao
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Curriculum Vitae

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Profile

Ph.D. candidate in experimental nuclear and particle physics developing reconstruction and inference methods for complex detector data. My research connects detector calibration, response simulation, inverse problems, statistical inference, and physics-informed machine learning across nuclear, heavy-ion, and neutrino experiments. I have published five first-author peer-reviewed articles.

Research Experience

Nucleon Short-Range Correlations

Oct. 2023 – Present

CSHINE Experiment, RIBLL-1 (HIRFL), Lanzhou, China

- Within the IBUU-MDI framework, constrained the SRC-induced high-momentum-tail fraction in ^{124}Sn to $R_{\text{HMT}} = (20 \pm 3)\%$ using precision hard- γ spectra and detector-filtered transport calculations.
- Developed calibration and detector-response models for the CsI(Tl) γ -ray hodoscope used in bremsstrahlung measurements from low-energy heavy-ion collisions.
- Implemented Richardson–Lucy spectrum unfolding and quantified reconstruction/systematic uncertainties through pseudo-experiments, closure tests, and consistency checks.

Correlation Function Imaging

Feb. 2024 – Present

Department of Physics, Tsinghua University

- Developed a Richardson–Lucy framework that directly images femtosopic source functions without imposing a Gaussian shape and simultaneously constrains strong-interaction parameters.
- Reconstructed non-Gaussian source functions from published pp and $\bar{p}p$ correlations in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV; the two source functions agree within uncertainty.
- Contributed to a published three-dimensional extension for pion-source imaging and neutron-skin sensitivity studies in heavy-ion collisions.

Deep-Sea Neutrino Detection

Jun. 2024 – Present

TRIDENT Collaboration; first-author manuscript prepared for JINST submission

- Built an atmospheric-neutrino simulation-to-reconstruction chain combining HONDA fluxes, FLUKA interactions, and Geant4 detector transport for a compact underwater detector concept motivated by TRIDENT.
- Developed a PyTorch Geometric graph-neural-network workflow using geometry-aware graph construction and Edge Convolution to reconstruct event topology, direction, and energy from sparse detector responses.
- Connected reconstructed outputs to oscillation-parameter inference over 64,273 selected events and recovered the input parameters within uncertainty through a physics-level closure test.

Experimental Experience

- **CSHINE Experiment** (Lanzhou, China, 2024): participated in beam-time data taking and led calibration, event reconstruction, and spectrum extraction for the high-energy γ -ray detector array.
- **SLEGS Beam Test at SSRF** (Shanghai, China, Jan. 2025): designed and led a dedicated CsI(Tl) response experiment from detector testing and setup through beam measurement, Geant4 calibration, and analysis.
- **S π RIT Experiment** (RIKEN, Japan, 2024): supported detector operation and electronics control during SAMURAI beam time, contributing to stable data taking and run coordination.

Technical Skills

Physics analysis	Detector calibration, detector-response simulation, event reconstruction, spectrum unfolding, systematic uncertainty evaluation.
Inference	Inverse problems, Richardson–Lucy deconvolution/imaging, likelihood inference, particle filtering, unscented Kalman filtering, pseudo-experiments, closure tests.
Machine learning	PyTorch, PyTorch Geometric, graph neural networks, Edge Convolution, LoRA/PEFT, physics-informed reconstruction.
Computing	Python, C/C++, ROOT, Geant4, FLUKA-based workflows, Linux, Git, HTCondor, HPC/GPU workflows, LaTeX.
Experiment	CAEN-based electronics operation, data acquisition, detector testing, beam-time operation, and run coordination.

Education

Tsinghua University

Aug. 2022 – Jun. 2027 (Expected)

Ph.D. Candidate in Physics; specialization in Experimental Nuclear and Particle Physics

Research areas: short-range correlations, heavy-ion collisions, femtoscopy, neutrino simulation, and machine-learning-based reconstruction.

South China Normal University

Aug. 2018 – Jun. 2022

B.S. in Optoelectronic Information Science and Engineering

GPA: 4.06/5.00; Rank: 1/137. Undergraduate thesis in DIS/SIDIS phenomenology: *A Study of the Nuclear Medium Effect of Hadron Fragmentation Function*.

Selected Publications

Five first-author peer-reviewed articles; complete publication record available on [Google Scholar](#).

- **Xu, J. et al.**, "Experimental study of bremsstrahlung γ -ray emission and short-range correlations in $^{124}\text{Sn}+^{124}\text{Sn}$ collisions at 25 MeV/nucleon," *Phys. Rev. C* **113**, 044613 (2026).
- **Xu, J. et al.**, "Precise measurement of short-range correlations in nuclei from bremsstrahlung gamma-ray emission in low-energy heavy-ion collisions," *Phys. Rev. Res.* **7**, 043174 (2025).
- **Xu, J. et al.**, "Linear response of CsI(Tl) crystal to energetic photons below 20 MeV," *Nucl. Instrum. Meth. A* **1080**, 170787 (2025).
- **Xu, J. et al.**, "Imaging Freeze-Out Sources and Extracting Strong Interaction Parameters in Relativistic Heavy-Ion Collisions," *Chin. Phys. Lett.* **42**, 031401 (2025).
- **Xu, J. et al.**, "Reconstruction of Bremsstrahlung γ -rays spectrum in heavy ion reactions with Richardson-Lucy algorithm," *Phys. Lett. B* **857**, 139009 (2024).

Selected Presentation

- **5th International Workshop on Quantitative Challenges in SRC and EMC-Effect Research** (LBNL, remote, Jun. 2026): "Experimental Study of Bremsstrahlung Gamma-Ray Emission and Short-Range Correlations in Low-Energy Heavy-Ion Collisions."

Mentoring and Teaching

- Mentored an undergraduate project on a plastic-scintillator cosmic-ray detector and an undergraduate thesis on GNN-based neutrino-event topology classification.
- Outstanding Teaching Assistant, Tsinghua University (2024).

Awards

- First-class Comprehensive Scholarship, Tsinghua University (2025).
- Second-class Comprehensive Scholarship, Tsinghua University (2024).
- Third Prize, National Undergraduate Mathematics Competition Final Round, Non-Math Major (2021).
- First Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM, 2020).